Digital Image Processing Lab (NCSH 302) **Session: Monsoon 2026-27**
B. Tech. 5th Sem (CSE-Hons) **Date: 29.09.2026**

Assignment-7 (Image Morphology)

1. Morphological image processing (or morphology) describes a range of image processing techniques that deal with the shape (or morphology) of features in an image. Morphological operations are typically applied to remove imperfections introduced during segmentation, and so typically operate on bi-level images. Erosion and dilation are basic morphological operations in image processing. Consider A as a binary image and B as a 5×5 square structuring element containing all 1's. Read the binary image *alphabet.png* and implement $erosion\,(A, B)$ and $dilation\,(A, B)$ to show the characteristics of erosion and dilation. Subsequently, perform boundary extraction on the binary image. Display the original image, eroded image, dilated image, and extracted boundary.

[15 Marks]

Sample Input:



Sample Output:



2. An inspection image *region_filling.png* contains several ring-shaped foreground objects. The ring structures are represented by foreground pixels with value 1, while the remaining pixels are background pixels with value 0. Each ring encloses a background region inside it; therefore, the image contains one external background region surrounding the objects and several enclosed background regions corresponding to the interiors of the rings. The objective is to first identify all background connected components using morphological operations, distinguish the enclosed background regions (holes) from the external background, and then fill each identified hole using morphological region filling. Use a 3×3 cross-shaped structuring element B for all morphological operations.

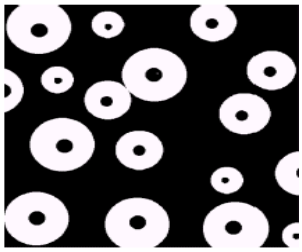
Part A - Morphological Connected-Component Extraction and Hole Identification: Obtain the background image using $A^c = 1 - A$. Starting from an unprocessed background pixel p_i, initialize $X_0^i = \{p_i\}$ and extract its complete connected background component using $X_k^i = \left(X_{k-1}^i \oplus B\right) \cap A^c$, iterating until $X_k^i = X_{k-1}^i$. Label the converged region as C_i, mark its pixels as processed, and select another unprocessed background pixel to extract the next component. Continue until all background pixels are assigned to components $C_1, C_2, \ldots, C_m$. Identify the component that touches the image boundary as the external background C_{ext}. Each remaining background component that does not touch the image boundary and is completely enclosed by a foreground ring must be labeled as a hole $H_1, H_2, \ldots, H_h$. Here, H_j represents the background region inside the j-th foreground ring, not the foreground ring itself. Display the original image, background image, external background component, and all identified hole regions with separate labels. Report initial seed position for each hole region.

[15 Marks]

Part B - Morphological Region Filling: Using the hole labels obtained in Part A, select one seed pixel p_j from inside each hole H_j. For each hole, initialize $Y_0^j = \{p_j\}$ and perform morphological region filling using $Y_k^j = \left(Y_{k-1}^j \oplus B\right) \cap A^c$ until $Y_k^j = Y_{k-1}^j$. The converged region Y^j should represent the complete background region enclosed by the corresponding foreground ring. Obtain the filled image for each hole using $R_j = A \cup Y^j$, and finally obtain the image containing all filled holes using $R = A \cup Y^1 \cup Y^2 \cup \cdots \cup Y^h$. For the first hole, display the seed and at least three intermediate iterations before convergence. Display the individual filled results and the final image. Report the total number of holes, the number of pixels in each hole, the number of iterations required for each hole to converge, and the total number of pixels added to the original foreground image.

[10 Marks]

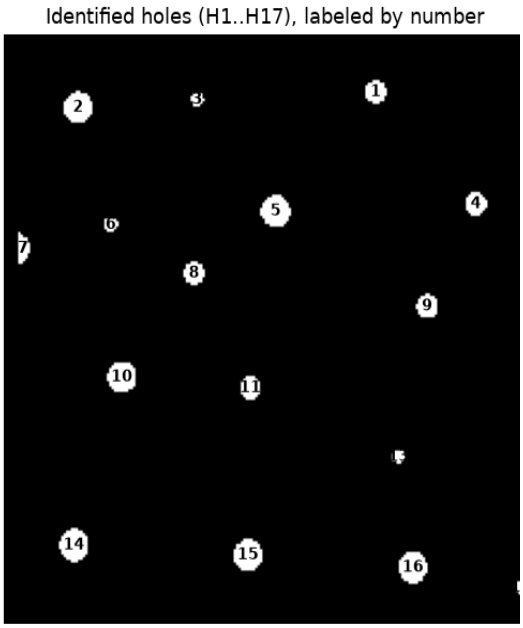
Sample Input:



Sample Output:

Part A:

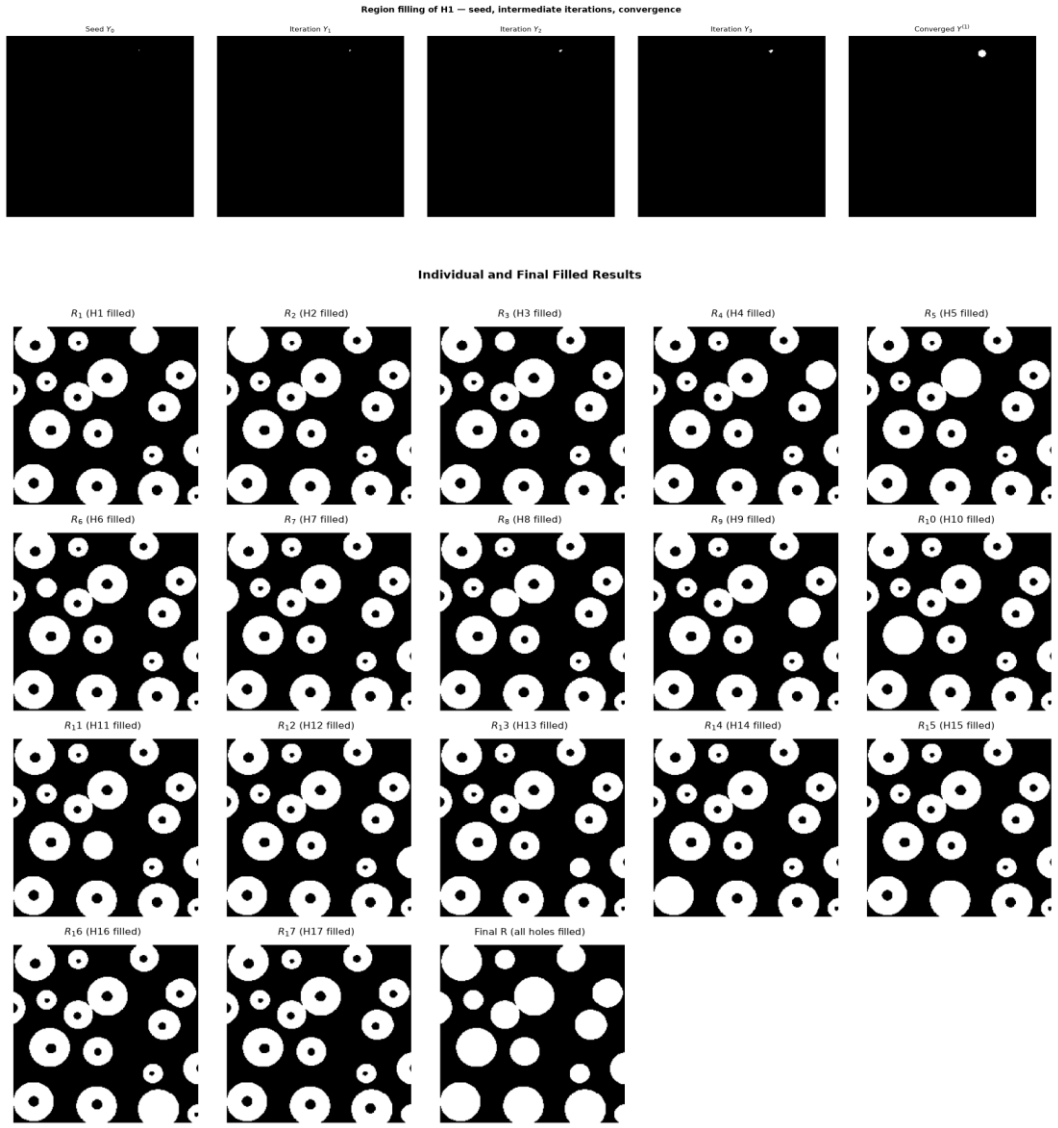




Total background components found: 18
Total holes identified: 17

H1: 63 pixels, seed = (17, 154)
H2: 112 pixels, seed = (21, 30)
H3: 22 pixels, seed = (22, 79)
H4: 63 pixels, seed = (59, 195)
H5: 117 pixels, seed = (60, 112)
H6: 25 pixels, seed = (69, 42)
H7: 44 pixels, seed = (74, 6)
H8: 61 pixels, seed = (85, 78)
H9: 64 pixels, seed = (97, 176)
H10: 111 pixels, seed = (123, 46)
H11: 60 pixels, seed = (128, 101)
H12: 12 pixels, seed = (148, 217)
H13: 23 pixels, seed = (156, 163)
H14: 118 pixels, seed = (185, 28)
H15: 115 pixels, seed = (189, 100)
H16: 114 pixels, seed = (194, 168)
H17: 18 pixels, seed = (205, 214)

Part B:



Total holes filled: 17

H1: 63 pixels, converged in 12 iterations
H2: 112 pixels, converged in 15 iterations
H3: 22 pixels, converged in 7 iterations
H4: 63 pixels, converged in 12 iterations
H5: 117 pixels, converged in 16 iterations
H6: 25 pixels, converged in 8 iterations
H7: 44 pixels, converged in 13 iterations
H8: 61 pixels, converged in 12 iterations
H9: 64 pixels, converged in 12 iterations
H10: 111 pixels, converged in 17 iterations
H11: 60 pixels, converged in 12 iterations
H12: 12 pixels, converged in 8 iterations
H13: 23 pixels, converged in 7 iterations
H14: 118 pixels, converged in 16 iterations
H15: 115 pixels, converged in 16 iterations
H16: 114 pixels, converged in 17 iterations
H17: 18 pixels, converged in 7 iterations
Total pixels added to A → R: 1142
|A| = 19094, |R| = 20236, |R| - |A| = 1142